

AI 자동설계 비교 리포트

WTNHJ02 · 케이스 8 · 기존설계 vs 자동설계(최단 / Stub+그룹패턴) — 길이·꺾임·그룹핑 Factor + 3D 뷰

1. 전체 집계

전략	성공	총길이(mm)	그룹핑 F(평균)
기존설계	151	900,150	0.563
최단(A*)	150/151	1,427,700	0.248
Stub+그룹패턴	148/151	1,298,641	0.447

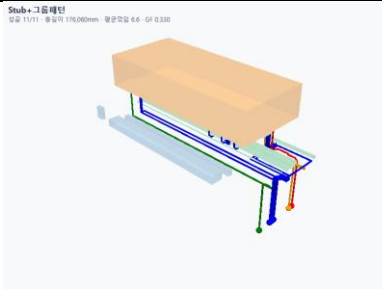
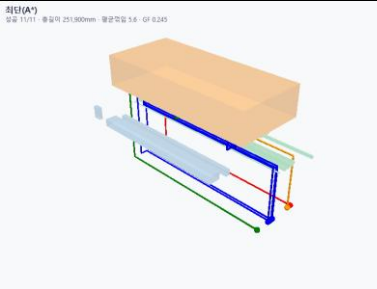
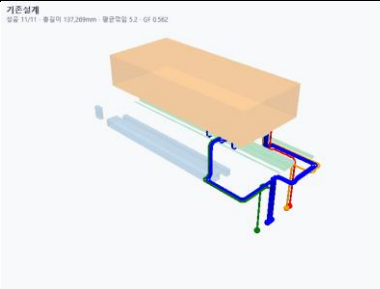
그룹핑 Factor = 0.35×랙집중도 + 0.30×번들밀집도 + 0.20×pitch 일관성 + 0.15×레인정렬도 (각 0~1, N/A 성분은 가중 재정규화). 최단=길이·직선성 우위, Stub+ 그룹=그룹핑 F(다발화)·사람설계 추종 우위가 기대값이다.

2. 케이스별 비교

■ 메인장비: WTNHJ02_

케이스 1. WTNHJ02_ / Chemical (작업 11)

전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	137,269	5.2	77.3	16.7	45.8	100.0	0.562
최단(A*)	11/11	251,900	5.6	0.1	5.5	39.0	100.0	0.245
Stub+그룹패턴	11/11	176,060	6.6	33.8	5.8	28.8	90.9	0.330



기존설계

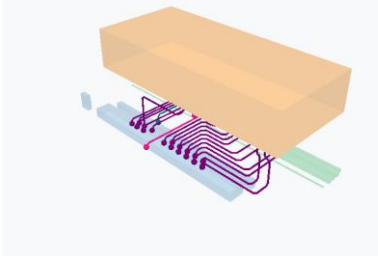
최단(A*)

Stub+그룹패턴

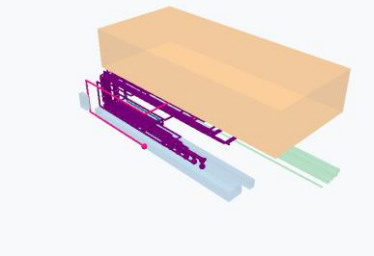
케이스 2. WTNHJ02_ / Exhaust (작업 20)

전략	성공	총길이(mm))	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	128,269	3.9	89.2	2.3	30.0	90.0	0.514
최단(A*)	20/20	316,100	7.5	5.4	5.8	38.9	90.0	0.249
Stub+그룹패턴	20/20	166,445	7.0	62.9	2.2	21.7	95.0	0.413

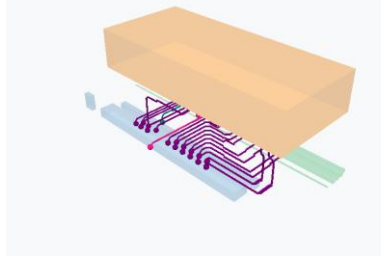
기존설계
성공 20/20 - 총길이 128,269mm - 평균꺾임 3.9 - GF 0.514



최단(A*)
성공 20/20 - 총길이 316,100mm - 평균꺾임 7.5 - GF 0.249



Stub+그룹패턴
성공 20/20 - 총길이 166,445mm - 평균꺾임 7.0 - GF 0.413



기존설계

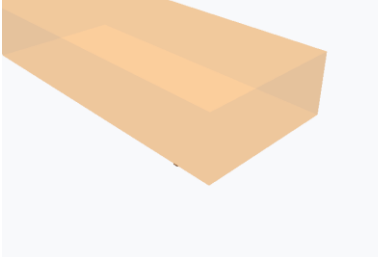
최단(A*)

Stub+그룹패턴

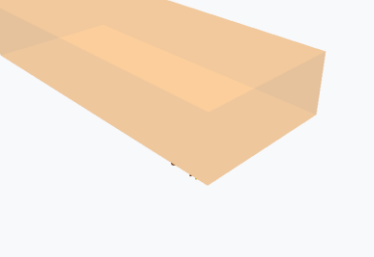
케이스 3. WTNHJ02_ / Gas (작업 33)

전략	성공	총길이(mm))	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	101,476	4.2	81.1	37.6	58.3	96.6	0.658
최단(A*)	33/33	82,200	2.9	37.1	8.8	28.5	100.0	0.363
Stub+그룹패턴	30/33	115,657	7.4	40.6	11.3	30.2	96.4	0.381

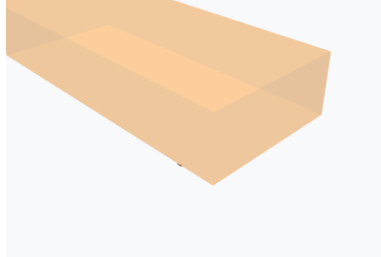
기존설계
성공 33/33 - 총길이 101,476mm - 평균꺾임 4.2 - GF 0.658



최단(A*)
성공 33/33 - 총길이 82,200mm - 평균꺾임 2.9 - GF 0.363



Stub+그룹패턴
성공 30/33 - 총길이 115,657mm - 평균꺾임 7.4 - GF 0.381



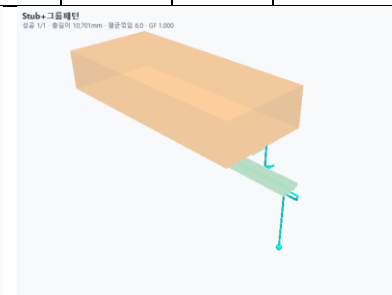
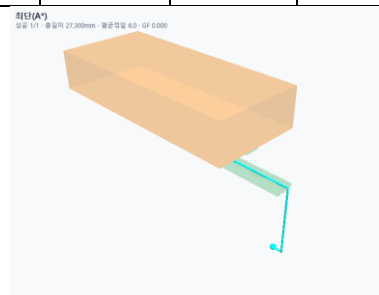
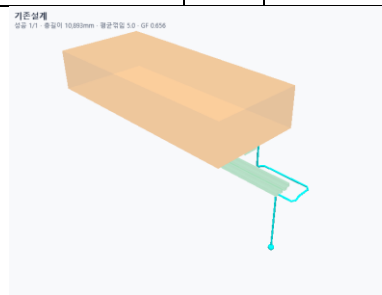
기존설계

최단(A*)

Stub+그룹패턴

케이스 4. WTNHJ02_ / Toxic (작업 1)

전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	10,893	5.0	65.6	N/A	N/A	N/A	0.656
최단(A*)	1/1	27,300	6.0	0.0	N/A	N/A	N/A	0.000
Stub+그룹패턴	1/1	10,701	6.0	100.0	N/A	N/A	N/A	1.000



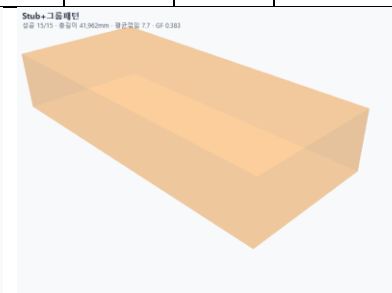
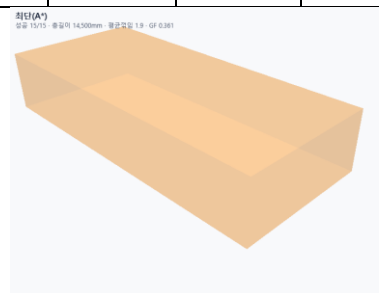
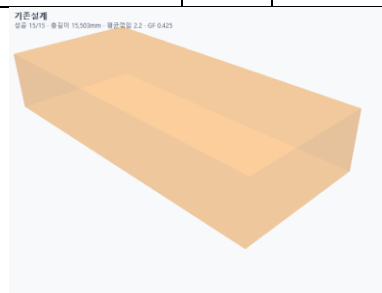
기존설계

최단(A*)

Stub+그룹패턴

케이스 5. WTNHJ02_ / UPW (작업 15)

전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	15,503	2.2	N/A	6.7	53.2	100.0	0.425
최단(A*)	15/15	14,500	1.9	N/A	7.2	31.4	100.0	0.361
Stub+그룹패턴	15/15	41,962	7.7	N/A	8.5	42.8	91.7	0.383



기존설계

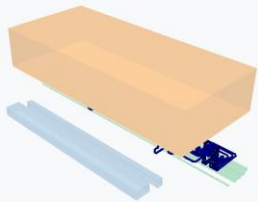
최단(A*)

Stub+그룹패턴

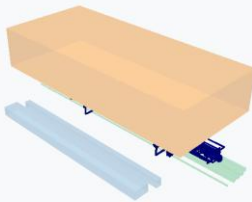
케이스 6. WTNHJ02_ / Waste Liquid (작업 40)

전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	316,291	7.1	68.4	15.5	20.8	97.5	0.474
최단(A*)	40/40	525,700	6.1	5.0	3.5	29.5	87.5	0.218
Stub+그룹패턴	40/40	558,826	9.8	12.5	3.5	21.4	92.5	0.236

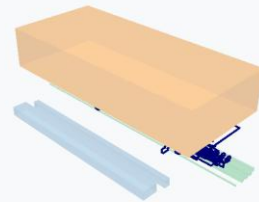
기존설계
성공 40/40 - 총길이 316,291mm - 평균꺾임 7.1 - GP 0.474



최단(A*)
성공 40/40 - 총길이 525,700mm - 평균꺾임 6.1 - GP 0.218



Stub+그룹패턴
성공 40/40 - 총길이 558,826mm - 평균꺾임 9.8 - GP 0.236



기존설계

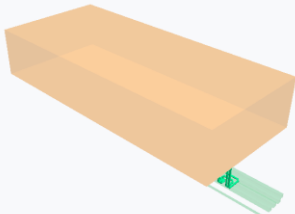
최단(A*)

Stub+그룹패턴

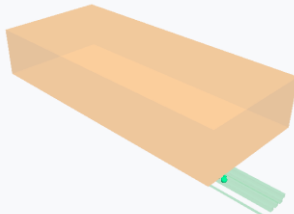
케이스 7. WTNHJ02_ / Waste Water (작업 11)

전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	85,459	6.5	91.1	10.8	35.5	100.0	0.572
최단(A*)	11/11	121,100	4.8	5.3	10.6	33.2	54.5	0.199
Stub+그룹패턴	11/11	113,853	8.3	47.3	8.6	51.2	80.0	0.414

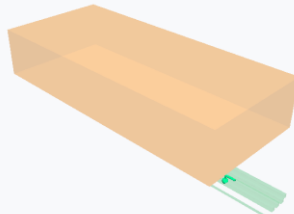
기존설계
성공 11/11 - 총길이 85,459mm - 평균꺾임 6.5 - GP 0.572



최단(A*)
성공 11/11 - 총길이 121,100mm - 평균꺾임 4.8 - GP 0.199



Stub+그룹패턴
성공 11/11 - 총길이 113,853mm - 평균꺾임 8.3 - GP 0.414



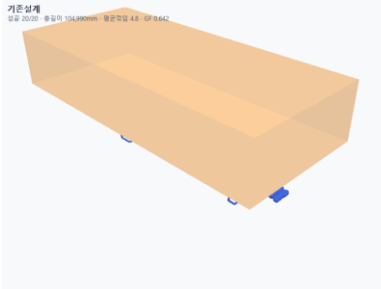
기존설계

최단(A*)

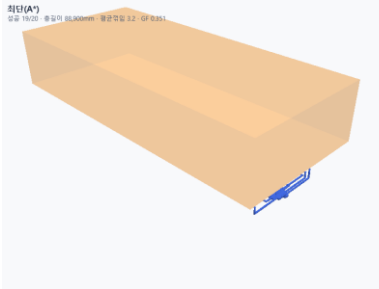
Stub+그룹패턴

케이스 8. WTNHJ02_ / Water (작업 20)

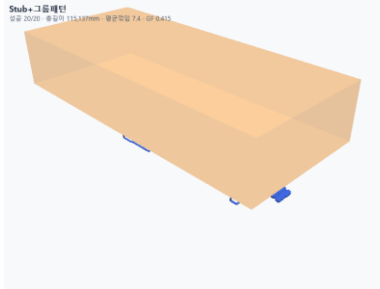
전략	성공	총길이(mm)	평균꺾임	랙집중 %	번들밀집 %	pitch %	lane %	그룹핑 F
기존설계	—	104,990	4.8	90.3	26.2	48.9	100.0	0.642
최단(A*)	19/20	88,900	3.2	38.9	7.8	32.7	84.2	0.351
Stub+그룹패턴	20/20	115,137	7.4	55.0	5.6	31.5	95.0	0.415



기존설계



최단(A*)



Stub+그룹패턴